

Exponential-Range Mass Production of Boolean Functions

Local recovery and optimal quantum state preparation

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Abstract

The hardest Boolean functions on n bits require $\Theta(2^n/n)$ gates to evaluate once. Must evaluating the same function on t unrelated inputs cost t times as much? We show that every Boolean function can be evaluated on any $t \leq 2^{\gamma n}$ inputs using a total of $O_\gamma(2^n/n)$ gates, for every fixed $0 \leq \gamma < 1$. Thus exponentially many evaluations fit within a constant factor of the worst-case cost of one. In the standard AND, OR, NOT basis, our bound is $(1/(1-\gamma) + o_\gamma(1))2^n/n$ gates. Uhlig previously achieved $(1 + o(1))2^n/n$ gates for the smaller range $t = 2^{o(n/\log n)}$.

Our construction encodes shorter restrictions of the function and recovers the requested outputs from disjoint sets of encoded functions. The gate bound includes choosing those sets, evaluating the functions, and routing the answers. The circuits are nonuniform; an efficient algorithm for constructing their fixed scheduling data remains open. A Lean companion verifies the Boolean upper bounds.

As an application, we determine the optimal worst-case cost of preparing t copies of a specified n -qubit pure state throughout the same range: $\Theta_\gamma((2^n/n) \log_2(t+1))$ elementary Clifford+ T gates. The error is at most $1/10$ in trace distance for the entire batch, and scratch qubits are restored to zero. Coherent Boolean evaluation gives the upper bound; a packing and circuit-counting argument gives the matching lower bound.

1 Introduction

How much circuit size can be saved when the same function must be evaluated on many unrelated inputs? This is a direct-sum question in which the outputs are independent but the circuit implementing them need not be: intermediate computations may be reorganized and shared across copies.

For a Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ and an integer $t \geq 1$, define

$$f^{\times t}(x^{(1)}, \dots, x^{(t)}) = (f(x^{(1)}), \dots, f(x^{(t)})).$$

The naive construction uses t disjoint copies of a circuit for f and gives

$$C(f^{\times t}) \leq tC(f).$$

Here C counts elementary Boolean gates. The classical Shannon–Lupanov bounds say that the hardest functions on n bits have circuit size $\Theta(2^n/n)$; in our basis the worst-case size is $(1 + o(1))2^n/n$

[Sha49, Lup58, FM05]. We call this the worst-case cost of one evaluation. The mass-production problem asks how large t can become before the cost must rise above that scale. For unrestricted circuits, the naive direct-sum bound can be far from optimal. Uhlig’s theorem allows $t = 2^{o(n/\log n)}$ independent evaluations with no asymptotic increase in the leading worst-case one-copy synthesis cost [Uhl92, Weg87, Kre23].

We prove, by a different construction, that the order-of-growth plateau reaches 2^γ simultaneous evaluations for every fixed $\gamma < 1$. The asymptotic coefficient is at most $1/(1 - \gamma)$ in our basis. This does not preserve Uhlig’s coefficient 1 for positive fixed γ .

1.1 Main result

We fix the standard basis $\{\wedge, \vee, \neg\}$, with fan-in two for the binary gates. Constant sources, fan-out, and output designations are free; circuit size counts the $\wedge$, $\vee$, and $\neg$ gates. Any other fixed finite complete basis changes the result by at most a constant-factor simulation.

Theorem 1.1 (Exponential-range mass production). *For every fixed $0 \leq \gamma < 1$, there is a constant A_γ such that, for every integer $n \geq 1$, every Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$, and every integer $1 \leq t \leq 2^\gamma$,*

$$C(f^{\times t}) \leq A_\gamma \frac{2^n}{n}.$$

Theorem 1.2 (Quantitative leading coefficient). *For every fixed $0 \leq \gamma < 1$, uniformly over Boolean functions on n bits and integers $1 \leq t \leq 2^\gamma$,*

$$C(f^{\times t}) \leq \left(\frac{1}{1 - \gamma} + o_\gamma(1) \right) \frac{2^n}{n}.$$

Equivalently, for every $\varepsilon > 0$, the coefficient $1/(1 - \gamma) + \varepsilon$ suffices for all sufficiently large n .

For fixed γ , the bound is independent of the batch size throughout the stated range. At the largest allowed batch size, the amortized size per requested value is therefore

$$\frac{C(f^{\times t})}{t} = O_\gamma \left(\frac{2^{(1-\gamma)n}}{n} \right).$$

This is a worst-case synthesis guarantee. For a particular function with a small circuit, the replication bound $tC(f)$ may be smaller; one may always take the better of the two bounds.

There must nevertheless be a transition before the 2^n -copy scale. If $n \geq 2$ and f depends on all n variables, every circuit for $f^{\times t}$ has size at least $t(n - 1)$. Hence a uniform $O(2^n/n)$ plateau cannot persist once $t = \omega(2^n/n^2)$; see Section 11. Our theorem reaches every fixed exponential rate, but leaves this near-endpoint regime open.

1.2 An optimal quantum state preparation tradeoff

Mass production also matters when many copies of the same quantum state are needed. A synthesis circuit may be designed for the specified state, and sharing can occur while the copies are prepared from all-zero inputs. Kretschmer established quantum analogues of Uhlig’s theorem [Kre23]; recent work connects mass production to coherent data loading and quantum algorithms [HKW25]. Our

Boolean theorem yields a matching upper and lower bound for approximate state preparation in a finite gate set.

Let $M(n, t)$ be the maximum, over n -qubit pure states $|\psi\rangle$, of the minimum number of elementary Clifford+ T gates needed to prepare $|\psi\rangle^{\otimes t}$ to trace-distance error at most $1/10$. The error is measured on all nt output qubits together: no measurement on the batch can change an outcome probability by more than $1/10$. Gates may connect any qubits; scratch qubits start and end in zero. We count every gate, including Clifford gates, and permit circuits tailored to ψ . [Section 10](#) gives the precise model and the proof.

Theorem 1.3 (Optimal approximate quantum state mass production). *For every fixed $0 \leq \gamma < 1$, uniformly over integers $1 \leq t \leq 2^{\gamma n}$ as $n \rightarrow \infty$,*

$$M(n, t) = \Theta_{\gamma} \left(\frac{2^n}{n} \log_2(t + 1) \right).$$

The cost grows logarithmically with the number of copies. In particular, it is $\Theta(2^n/n)$ for one copy, $\Theta(2^n \log n/n)$ for $t = n^a$ with fixed $a > 0$, and $\Theta_{\gamma}(2^n)$ for $t = \lfloor 2^{\gamma n} \rfloor$ with fixed $0 < \gamma < 1$. The single-copy endpoint is already known [[GKW26](#)]. Our upper bound uses coherent Boolean mass production and carefully allocated rotation precision. The lower bound reflects the fact that t copies distinguish states whose single-copy distance is on the order of $1/\sqrt{t}$. The quantum arguments are written proofs and are outside the Lean companion.

1.3 How the construction works

Split an input as $x = (a, z)$, with a p -bit prefix a and a d -bit suffix z , where $p + d = n$. The prefix selects one of the 2^p shorter functions $f_a(z) = f(a, z)$. The desired saving would come from using only one circuit for each shorter function. But two inputs with the same prefix may require that function at two different suffixes, and one circuit has only one set of input wires.

Coding supplies alternative ways to answer each request. We replace the restrictions f_a by a larger bank of shorter functions, called *resources*. Each desired value can be recovered from many different sets of resources. A scheduler chooses one set for each request so that no resource is needed twice. Each resource circuit can then receive the one suffix assigned to it, and all requested answers can be reconstructed.

Uhlig serves two requests using opposite ends of an XOR chain, then scales through recursion [[Uhl92](#), [Weg87](#)]. We replace that chain with a code offering many recovery choices, and schedule disjoint choices for an entire batch. The key is making the scheduling cheap enough to preserve the savings. [Section 4](#) develops this comparison through a worked example.

The construction has four components.

1. **Pack restrictions.** For every fixed suffix z , encode the values $f(a, z)$ over all prefixes a in a code over $\mathbb{F}_q$. As z varies, each bit of each code coordinate defines a resource function on d variables.
2. **Recover locally.** The code coordinates are points in $\mathbb{F}_q^{\ell}$. A requested symbol is the sum of the $q - 1$ other symbols on any line through its information point. About $q^{\ell-1}$ directions are available, giving many recovery options.

3. **Schedule disjoint recovery.** Test a fixed menu of direction assignments and choose one that serves at least half the pending requests without collisions. Reserve those recovery sets and repeat on the remaining requests. The total gate cost is $\tilde{O}_\ell(tq)$, including selection of a schedule for the live inputs.
4. **Evaluate the resources.** Each code coordinate is now queried on at most one suffix. Apply ordinary one-copy synthesis to each resource function on $d = |z|$ variables, route its value to the requesting line, and decode.

The encoding defines the resource truth tables when the circuit is built. At evaluation time, the circuit selects recovery sets, routes suffixes, evaluates the fixed resource circuits, and decodes. The menus are fixed in advance and work for every batch; the input prefixes determine which entries the circuit selects.

1.4 Where the leading coefficient comes from

The main cost is the product of the number of resource functions and the cost of synthesizing one of them. A code whose rate tends to one, together with packing that leaves only a vanishing fraction of its capacity unused, gives

$$\underbrace{(1 + o(1))2^p}_{\text{Boolean resources}} \cdot \underbrace{(1 + o(1))\frac{2^d}{d}}_{\text{gates per resource}} = (1 + o(1))\frac{2^n}{d}.$$

Leaving more suffix bits makes this product smaller, but leaves fewer prefix bits to provide distinct recovery options. The scheduler can accommodate $t \leq 2^{\gamma n}$ while leaving $d = \delta n + O(1)$ suffix bits for any fixed $0 < \delta < 1 - \gamma$, by taking the geometric dimension ℓ sufficiently large and then fixed. Its scheduling and routing costs are exponentially smaller than $2^n/n$. The displayed product therefore gives the coefficient $1/\delta$. Choosing δ close enough to $1 - \gamma$ gives $1/(1 - \gamma) + \varepsilon$ for any fixed $\varepsilon > 0$. For example, the bound for $t \leq 2^{n/2}$ has coefficient $2 + o(1)$.

Machine-checked companion. An accompanying Lean 4 development verifies the Boolean recursive construction in a rational, discrete-exponent formulation and the complete nonuniform construction with the real-rate and additive-error quantifiers of [Theorem 1.2](#). [Appendix C](#) records the theorem endpoints, pinned sources, and finite accounting.

Development methodology. Generative-AI systems materially assisted mathematical experiments, literature discovery, and proof development, as detailed in the AI Disclosure at the end of the paper.

Organization. [Section 2](#) compares prior work before the constructions. [Section 4](#) illustrates disjoint recovery with two requests. [Sections 5 to 7](#) develop the code, scheduler, and composition bound used in the direct proof in [Section 8](#); [Section 9](#) sharpens the coefficient. [Section 10](#) proves the quantum state preparation tradeoff. An alternative proof in [Appendix A](#) uses an explicitly constructible greedy scheduler. Its quadratic scheduling cost is controlled by grouping requests and recursively handling resource reuse. [Appendices B and C](#) give circuit implementation details and the exact finite accounting used in Lean.

2 Relation to earlier work

The table compares the circuit bounds and the main coding antecedents. Here “one-copy leading term” means $(1 + o(1))2^n/n$ gates in our basis, the worst-case cost of evaluating one function once. Uhlig’s 1974 paper [Uhl74] is a precursor to the mass-production theorem presented in [Uhl92] and in Wegener’s Theorem 10.2.3 [Weg87].

Result	Copy range	Circuit-size bound	Comparison
Uhlig [Uhl92, Weg87, Kre23]	$\log t = o(n/\log n)$	one-copy leading term	optimal order and leading constant in a smaller copy range
Paul, Lemma 2 [Pau76]	arbitrary t	$O(\max\{n2^n, nt \log^2 t\})$	$O(n2^n)$ throughout every fixed exponential range below one
Holmgren–Rothblum [HR24]	arbitrary t	$O(2^n + tn^3)$ after hardwiring	$O(2^n)$ throughout every fixed exponential range below one
Albrecht, Theorem 1 [Alb89]	$t \leq 2^n$	$O(2^n n^2)$	the full exponential range, with polynomial overhead
Finite-geometry batch codes [IKOS04, PV19]	code-dependent batch size	disjoint local recovery; no bound on $C(f^{\times t})$	coding antecedents for affine-geometric recovery
This paper	$t \leq 2^{\gamma n}$ for fixed $\gamma < 1$	$O_\gamma(2^n/n)$	optimal order; limiting coefficient at most $1/(1 - \gamma)$

Paul’s construction treats the truth table as an ordered database: sort the input addresses, sweep through the hardwired table, and restore the answers to their original order. For fixed $\gamma < 1$ and $t \leq 2^{\gamma n}$, $nt \log^2 t = o(n2^n)$, giving the table’s $O(n2^n)$ bound. Paul’s g^k is the same product task considered here; his arbitrary binary Boolean gates can be simulated in our basis at constant cost.

Holmgren and Rothblum prove that Q indexed bits can be selected from a live N -bit database by a bounded-fan-in Boolean circuit of size $O(N + Q \log^3 N)$ and depth $O(\log(N + Q))$ [HR24, Main theorem]. In our nonuniform model, hardwiring the database to the 2^n -bit truth table of f and using the t live n -bit input blocks as indices gives $O(2^n + tn^3)$, as recorded in the table. Thus, throughout every fixed exponential range below one, Theorem 1.1 improves the bounds from Paul’s construction and this multiselection reduction by factors of n^2 and n , respectively. Multiselection also implements the gather step of our routing; Appendix B.3 gives the reduction.

Albrecht studies the same simultaneous-realization problem over a complete binary basis [Alb89, Theorem 1, pp. 53–54]. His construction covers the endpoint $t = 2^n$, with depth $O(n \log n)$ and size a factor $O(n^3)$ above the one-copy scale, as Albrecht also notes.

Hiltgen and Paterson use “mass production” for PI_k -set circuits underlying monotone slice functions [HP95]. Their result gives sharing in that restricted set-circuit setting and leads to an optimal circuit for the Nečiporuk slice. The task here is simultaneous evaluation of an arbitrary Boolean function on disjoint input blocks.

Coding ingredients and circuit costs. Systematic polynomial evaluation, affine-line recovery, and disjoint recovery sets are established batch-code tools. Section 3.4 of Ishai–Kushilevitz–

Ostrovsky–Sahai [IKOS04] chooses random affine lines and deletes intersections while preserving enough points for interpolation. Their Remark 3.11 also observes that the randomized decoding algorithms in that and the following sections can be derandomized using limited independence. In their terminology, the repeated-request analogue relevant here is a primitive multiset batch code. Our local gadget uses the elementary tensor-product evaluation subspace, with a separate degree bound in each variable, whereas their construction uses total-degree Reed–Muller evaluation.

Finite-geometry constructions of primitive batch codes likewise give explicit families of mutually disjoint recovery sets [PV19]. Later lifted-polynomial constructions also use line-based recovery to obtain batch-code guarantees [HPPV20]. Private-information-retrieval codes of Fazeli, Vardy, and Yaakobi form a related family in which each information symbol has many disjoint recovery options [FVY15].

For circuit mass production, selecting recovery sets is itself part of the computation: the prefixes are input wires, and every batch must produce valid recovery records within the gate budget. A generic polynomial bound in the exponentially large code length and batch size need not fit below $2^n/n$. The menu scheduler provides the more precise bound $\bar{O}_\ell(tq)$, which Proposition 7.1 combines with resource synthesis and routing. Counting occupied-line descriptions keeps the menus short; sharing the occupied-point list across candidates keeps their verification small. Together with high-rate packing, this gives the stated exponential-range bound and its leading coefficient.

Quantum mass production and synthesis. Kretschmer [Kre23] gives exact synthesis of $t = 2^{o(n/\log n)}$ copies of any n -qubit state with $O(2^n)$ CNOT gates, allowing arbitrary one-qubit gates. He also obtains an $O(4^n)$ bound for t uses of an arbitrary n -qubit unitary. His discussion asks what happens for larger copy ranges. Theorem 1.3 treats a finite gate set and constant error for the whole batch; these model differences are essential to the comparison with exact synthesis.

Huggins, Khattar, and Wiebe [HKW25] combine mass production with quantum read-only memory, which loads fixed classical data at addresses held in quantum registers. They give applications to quantum algorithms and analyze both a small-copy plateau and a larger-copy regime. Their Appendix B.5 explains the passage from data loading to state preparation. Our upper proof uses this established connection with the stronger Boolean synthesis bound.

Gosset, Kothari, and Wu [GKW26] determine the optimal T -gate count for state preparation and diagonal-unitary synthesis. Their discussion also records an $O(2^n/n)$ total-gate bound for preparing one state at constant error, and their Appendix B allocates approximation error geometrically across conditional rotations. We count all elementary gates and determine their dependence on t through an exponential copy range. Our proof uses standard conditional-rotation synthesis [SBM06] and finite-gate approximation [DN06], followed by a matching packing lower bound.

3 Circuit conventions and worst-case notation

Our circuits are finite directed acyclic graphs. Source vertices are input bits or the constants zero and one. Internal vertices are labelled by $\wedge$, $\vee$, or $\neg$, with the usual fan-in. A circuit for a map $F : \{0, 1\}^N \rightarrow \{0, 1\}^m$ has m designated output wires, which may be input wires or gate outputs and may repeat. Fan-out, constants, and output designations carry no cost; size is the number of internal gates. We write $C(F)$ for the minimum size of such a circuit. All leading-constant comparisons in this paper refer to this fixed model.

Let

$$L(n) = \max_{f: \{0,1\}^n \rightarrow \{0,1\}} C(f).$$

The Shannon counting lower bound and Lupanov's synthesis upper bound imply

$$L(n) = \Theta(2^n/n)$$

for every fixed finite complete basis; see [Sha49, Lup58, FM05, Weg87]. We use only the upper bound.

Fixing all but one input block and retaining its output gives $C(f^{\times t}) \geq C(f)$. Thus $2^n/n$ is the optimal worst-case order even when several outputs are requested.

All circuits below are nonuniform. Field representations, irreducible polynomials, sorting networks, and other finite combinatorial objects may be fixed as part of the circuit for each input length.

The main parameters have the following roles. Code length and dimension are measured in field symbols; multiplying by b converts either count to bits.

Parameter	Meaning
$n = p + d, t$	Input length, prefix/suffix split, and number of requests
$q = 2^b, \ell$	Field size, bits per symbol, and fixed geometric dimension
$N = q^\ell, K$	Code length and dimension; K/N is the code rate
$D_q = (q^\ell - 1)/(q - 1)$	Number of line directions through each target
g	Number of requests in the scheduler lemmas
M	Number of slots in a routing table

We use $\tilde{O}_\ell(X)$ for

$$X \cdot \text{poly}_\ell(n, \log q, \log(t + 2), \log(M + 2)).$$

Only polynomial factors are suppressed; in the parameter regimes used below, they never affect an exponential rate.

For a positive integer k , write $[k]_0 = \{0, \dots, k - 1\}$.

We repeatedly use two standard circuit primitives. First, after representing $\mathbb{F}_{2^b}$ in a polynomial basis, field addition, multiplication, inversion of a nonzero element, and comparison of canonical b -bit representations have circuits of size $\text{poly}(b)$. For example, inversion can be implemented by binary exponentiation to the power $2^b - 2$. Second, R records of width w can be sorted by a fixed sorting network using $O(R \log^2 R)$ compare-exchange units, each of size $O(w)$; see [Bat68]. Sorting by a leading type bit, with original positions used as tie breakers when needed, also gives stable fixed-wire compaction within the same asymptotic bound.

4 The two-copy construction as disjoint recovery

Uhlig's two-copy argument illustrates the invariant behind disjoint recovery: each requested value has two representations, and the requests can choose representations that use separate resources.

Fix a prefix length p , put $M = 2^p$, and define the restrictions

$$f_i(z) = f(i, z), \quad i \in \{0, \dots, M - 1\}.$$

Uhlig's two-copy construction introduces the following functions; see the proof of Wegener's Theorem 10.2.3 [Weg87] and the overview in [Kre23, Section 1.2]:

$$g_0 = f_0, \quad g_i = f_{i-1} \oplus f_i \quad (1 \leq i < M), \quad g_M = f_{M-1}.$$

Every restriction has two representations:

$$f_i = \bigoplus_{j=0}^i g_j = \bigoplus_{j=i+1}^M g_j.$$

Given two inputs $(i, z^{(1)})$ and $(j, z^{(2)})$ with $i \leq j$, recover $f_i(z^{(1)})$ from the prefix set $\{0, \dots, i\}$ and $f_j(z^{(2)})$ from the suffix set $\{j+1, \dots, M\}$. The sets are disjoint, so each resource function g_j is evaluated on at most one suffix.

For example, take $p = 2$, so there are four restrictions and five resources. The requests with prefixes 1 and 2 can be answered as

$$f_1(z^{(1)}) = g_0(z^{(1)}) \oplus g_1(z^{(1)}), \quad f_2(z^{(2)}) = g_3(z^{(2)}) \oplus g_4(z^{(2)}).$$

Resources g_0, g_1 receive $z^{(1)}$, resources g_3, g_4 receive $z^{(2)}$, and g_2 is unused. The suffixes can be completely unrelated. Even if the two prefixes coincide, the prefix and suffix recovery sets remain disjoint.

For two requests arriving in the opposite order, a comparator first orders their prefixes, conditional swaps route the two suffixes to the prefix and suffix recovery branches, and a final swap restores the output order. This costs only $\text{poly}(n)$ gates.

The map $(f_0, \dots, f_{M-1}) \mapsto (g_0, \dots, g_M)$ is, up to an invertible change of information coordinates, the length- $(M+1)$ single-parity-check code. The two representations above are disjoint recovery sets for each information coordinate. To serve exponentially many requests, we use a constant-rate code with many geometric recovery sets.

5 The local-recovery gadget

We need a code with three features: it stores the 2^p prefix values using only $O(2^p)$ bits, it offers many recovery sets for every value, and the circuit can compute the addresses in those sets cheaply. The following product code supplies all three. [Proposition 5.2](#) is the interface used by the rest of the proof; the polynomial construction establishes that interface.

Fix an integer $\ell \geq 2$ and a field $\mathbb{F}_q$ of characteristic two, where $q = 2^b$. Set

$$s_0 = \left\lfloor \frac{q-1}{\ell} \right\rfloor$$

and fix a polynomial-basis representation of $\mathbb{F}_q$. If α_i denotes the field element whose canonical b -bit representation is the integer i , choose the efficiently indexed set

$$A = \{\alpha_0, \dots, \alpha_{s_0-1}\}.$$

Let $\mathcal{V}$ be the space of polynomials

$$P(X_1, \dots, X_\ell) \in \mathbb{F}_q[X_1, \dots, X_\ell]$$

having degree less than s_0 in every variable. Repeated univariate interpolation shows that evaluation on A^ℓ uniquely determines P , and

$$\deg P \leq \ell(s_0 - 1) \leq q - 2.$$

The evaluation map

$$\text{Enc} : \mathbb{F}_q^{A^\ell} \longrightarrow \mathbb{F}_q^\ell, \quad (P(u))_{u \in A^\ell} \longmapsto (P(y))_{y \in \mathbb{F}_q^\ell},$$

is a systematic linear code: the original information symbols appear unchanged at the coordinates in A^ℓ . Its length and dimension are

$$N = q^\ell, \quad K = s_0^\ell = \Theta_\ell(q^\ell).$$

5.1 Packing Boolean restrictions

Split an n -bit input as

$$x = (a, z), \quad |a| = p, \quad |z| = d = n - p.$$

For each fixed suffix z , pack the 2^p bits

$$(f(a, z))_{a \in \{0,1\}^p}$$

into K field symbols, using the fixed $\mathbb{F}_2$ -basis of $\mathbb{F}_q$ and padding unused bit positions with zeros. More explicitly, interpret a prefix a as an integer $I(a) \in [2^p]_0$, and set

$$s(a) = \left\lfloor \frac{I(a)}{b} \right\rfloor, \quad j(a) = I(a) \bmod b.$$

Write $s(a)$ in base s_0 using exactly ℓ digits $i_1(a), \dots, i_\ell(a) \in [s_0]_0$, and define

$$u(a) = (\alpha_{i_1(a)}, \dots, \alpha_{i_\ell(a)}) \in A^\ell.$$

Thus bit $j(a)$ of the information symbol at $u(a)$ is $f(a, z)$. Fixed-divisor division, remainder, and base conversion have Boolean circuits of size $\text{poly}_\ell(p, \log q)$, so computing $(u(a), j(a))$ for all requests costs $\tilde{O}_\ell(t)$ gates.

Choose q , among powers of two, minimally so that

$$Kb \geq 2^p.$$

For all sufficiently large p (depending on ℓ), the corresponding code dimensions at field sizes q and $q/2$ are both $\Theta_\ell(q^\ell)$. Minimality then gives

$$q^\ell b = \Theta_\ell(2^p). \tag{1}$$

Indeed, the lower bound follows from $Kb \leq q^\ell b$. For the upper bound, the preceding candidate $q/2$ has bit width $b-1$, fails the packing inequality, and has dimension $\Omega_\ell(q^\ell)$. Hence $q^\ell(b-1) = O_\ell(2^p)$; since $b \leq 2(b-1)$ for $b \geq 2$, the upper bound in (1) follows. Equivalently,

$$\ell \log_2 q + \log_2 \log_2 q = p + O_\ell(1), \quad \log_2 q = \frac{p}{\ell} + O_\ell(\log p). \quad (2)$$

Let P_z be the polynomial whose information symbols are the packed bits. For each code coordinate $y \in \mathbb{F}_q^\ell$, define the field-valued resource

$$G_y(z) = P_z(y).$$

Each of the b coordinate bits of G_y is a Boolean function on d variables, determined by f , y , and the chosen field basis. For $r \in [b]_0$, write

$$H_{y,r}(z) = (G_y(z))_r.$$

These definitions view the same array in two ways:

Held fixed	What varies	What the values describe
Suffix z	Code coordinate y	One codeword P_z
Code coordinate y	Suffix z	One resource function G_y

The decoder combines coordinates of a codeword. The resource circuit evaluates one column of this array at the suffix routed to it. No encoding circuit or full array is evaluated on the live inputs: the array specifies which shorter functions are synthesized.

5.2 Affine-line recovery

Lemma 5.1 (Line identity). *Let $P : \mathbb{F}_q^\ell \rightarrow \mathbb{F}_q$ be represented by a polynomial of total degree at most $q-2$. For every $u \in \mathbb{F}_q^\ell$ and every nonzero direction $v \in \mathbb{F}_q^\ell$,*

$$P(u) = - \sum_{\lambda \in \mathbb{F}_q^*} P(u + \lambda v).$$

In characteristic two, the minus sign may be omitted.

Proof. The restriction $Q(\lambda) = P(u + \lambda v)$ is a univariate polynomial of degree at most $q-2$. The sum of the constant monomial over $\mathbb{F}_q$ is $q \cdot 1 = 0$ in characteristic two. For $1 \leq j \leq q-2$, the sum of λ^j over $\mathbb{F}_q^*$ vanishes because j is not divisible by $q-1$, and the zero term contributes nothing. Applying these identities monomial by monomial gives $\sum_{\lambda \in \mathbb{F}_q} Q(\lambda) = 0$. Separating the term $\lambda = 0$ proves the claim. $\square$

Thus every affine line through an information point u supplies a recovery set consisting of its $q-1$ non-target points. We identify two nonzero direction vectors when one is a nonzero scalar multiple of the other and write $[v]$ for the resulting projective direction. The number of such directions through a point is

$$D_q = \frac{q^\ell - 1}{q - 1} = 1 + q + \cdots + q^{\ell-1}, \quad q^{\ell-1} \leq D_q < 2q^{\ell-1}. \quad (3)$$

For fixed ℓ and $p \rightarrow \infty$, Equations (2) and (3) give the parameter dictionary

$$q^\ell = 2^{p+o(p)}, \quad q = 2^{p/\ell+o(p)}, \quad D_q = q^{\ell-1+o(1)} = 2^{((\ell-1)/\ell)p+o(p)}. \quad (4)$$

Two distinct lines through the same target intersect only at that target, which is absent from both recovery sets. Repeated requests for one information symbol therefore cause no additional difficulty. A later target may itself belong to the occupied set, but it blocks no direction because every new recovery set omits its own target.

The facts needed later can now be packaged without reference to the internal polynomial representation.

Proposition 5.2 (Local-recovery gadget). *Fix $\ell \geq 2$ and a split $n = p + d$, with p sufficiently large depending on ℓ , and choose $q = 2^b$ by the packing rule above. For every $f : \{0, 1\}^n \rightarrow \{0, 1\}$ there are $q^\ell b$ Boolean resource functions $H_{y,r} : \{0, 1\}^d \rightarrow \{0, 1\}$ with the following properties.*

- (i) *The resource count satisfies $q^\ell b = \Theta_\ell(2^p)$.*
- (ii) *A prefix a determines an information point $u(a) \in A^\ell$ and a bit position $j(a) \in [b]_0$ by a circuit of size $\text{poly}_\ell(p, \log q)$.*
- (iii) *For every projective direction $[v]$, the set*

$$R(u(a), [v]) = \{u(a) + \lambda v : \lambda \in \mathbb{F}_q^*\}$$

has $q - 1$ coordinates and recovers the requested bit by

$$f(a, z) = \bigoplus_{y \in R(u(a), [v])} H_{y, j(a)}(z).$$

There are $D_q = (q^\ell - 1)/(q - 1)$ such recovery sets.

- (iv) *If $U \subseteq \mathbb{F}_q^\ell$ is already occupied, at most $|U \setminus \{u(a)\}|$ directions have $R(u(a), [v]) \cap U \neq \emptyset$.*

Proof. The packing construction gives (i) and (ii). The line identity, followed by selection of coordinate bit $j(a)$ in the fixed $\mathbb{F}_2$ basis, gives (iii). For (iv), every $y \in U \setminus \{u(a)\}$ lies on exactly one projective line through $u(a)$, while the omitted target $u(a)$ blocks none. $\square$

Thus one desired bit has become $q - 1$ requests to shorter resource functions, but those coordinates may be chosen from any of D_q recovery sets. The next section turns the blocking guarantee in (iv) into a deterministic circuit and then routes only the coordinates actually selected.

6 Deterministic scheduling and routing

The scheduler maps target coordinates to disjoint recovery sets. A greedy construction first shows how the blocking bound gives a circuit of size $\tilde{O}_\ell(g^2 q)$ for g requests. Fixed menus reduce this cost to $\tilde{O}_\ell(gq)$. Routing then connects the selected addresses to the resource circuits.

6.1 Greedy line scheduling

The geometric count says that a suitable line exists, but a circuit must find one from the target points. We do this greedily. Previously occupied points rule out directions; projective ranking and sorting let an ordinary Boolean circuit choose the first direction that remains.

Consider a multiset of g requested information points $u_1, \dots, u_g \in A^\ell$. Process the requests in order. Suppose disjoint recovery sets have been selected for requests $1, \dots, j-1$, and let U be their union. Then

$$|U| = (j-1)(q-1) < gq.$$

For the target u_j , every point $y \in U \setminus \{u_j\}$ forbids at most one projective direction, namely the direction of $y - u_j$. The point u_j itself forbids none because it is omitted from the new recovery set.

Lemma 6.1 (Projective indexing). *Projective directions in $\mathbb{F}_q^\ell$ have rank and unrank circuits of size $\text{poly}_\ell(b)$, with ranks in $[D_q]_0$.*

Implementation idea. The indexing normalizes the first nonzero coordinate to one and orders the normalized vectors by the position of that coordinate and then by their remaining base- q digits. The explicit circuits are given in [Appendix B.1](#).

Lemma 6.2 (Greedy line scheduler). *If*

$$gq < D_q, \tag{5}$$

then every multiset of g target points admits pairwise disjoint affine-line recovery sets. Given the targets, the recovery sets can be selected by a Boolean circuit of size

$$\tilde{O}_\ell(g^2q).$$

Proof idea. At stage j , the blocking property in [Proposition 5.2](#) leaves a valid direction because fewer than D_q directions are forbidden. The circuit ranks the forbidden directions, sorts those ranks, and selects the least missing one; stage j costs $\tilde{O}_\ell(jq)$. The treatment of duplicate targets, sentinel ranks, and the least-missing-rank circuit is given in [Appendix B.2](#). Summing over the stages gives the stated cost.

6.2 A nonuniform scheduler with linear dependence on the batch size

The quadratic scan can be replaced by phases that accept half the remaining requests. We prove that a small fixed menu of candidate directions works for every possible phase state, then give a circuit that tests the menu. The existence proof uses randomness only to choose constants of the circuit; its evaluation is deterministic on every input.

There are three steps. First, one random direction assignment succeeds with very high probability for any fixed state. Second, a union bound supplies a short menu that contains a successful assignment for every state. Third, a sorting circuit finds such an assignment without multiplying the occupied-set cost by the menu length. Here g is the original batch size, k is the number of pending requests, and U contains the points reserved earlier. A menu entry is a tuple of k directions, one for each pending request.

Lemma 6.3 (A partial schedule with exponentially small failure). *Put $N = q^\ell$. Suppose $512gq \leq D_q$, let $1 \leq k \leq g$, and let U be the union of at most g affine-line recovery sets. Fix any k targets, and choose one projective direction independently and uniformly for each target. Call a candidate recovery set clean if it avoids U and every other candidate recovery set. Then*

$$\Pr\{\text{fewer than } \lceil k/2 \rceil \text{ sets are clean}\} \leq 2^{-k}.$$

Proof. For any fixed set B , a random recovery set through any fixed target meets B with probability at most $|B|/D_q$: each point other than the target blocks exactly one direction. Consequently, the probability of meeting U is at most $a_0 = g(q-1)/D_q$, and that of meeting another fixed candidate recovery set is at most $a_1 = (q-1)/D_q$.

Pairwise collision events are dependent, so we cannot multiply these bounds directly. Instead, we extract a set of bad requests whose collision witnesses all lie outside that set, and condition on the outside directions.

Form a graph on $\{0, 1, \dots, k\}$. Join a request to vertex 0 when its recovery set meets U , and join two requests when their recovery sets meet. The nonclean requests are exactly the nonisolated vertices other than 0. Take a spanning forest of the nontrivial components and two-color it. Every nonclean request has a forest neighbor of the opposite color. If at least $\lceil k/2 \rceil$ requests are nonclean, one color contains at least $h = \lceil k/4 \rceil$ of them. Choose exactly h such requests as a set S , excluding vertex 0. Every request in S then has a collision witness outside S , either vertex 0 or an unselected request.

For each fixed set S of size h , condition on all directions outside S . Their recovery sets, together with U , form a fixed set of at most $(g+k)(q-1)$ points. Each selected request meets that set with probability at most $a_0 + ka_1$. These h tests are conditionally independent, since each uses only its own unexposed direction. Averaging over the complementary directions and taking a union bound over S gives

$$\Pr\{\text{such a collision cut exists}\} \leq \binom{k}{h} (a_0 + ka_1)^h \leq 2^k (1/256)^{k/4} = 2^{-k}.$$

Here $a_0 + ka_1 \leq 2gq/D_q \leq 1/256$. Failure of the stated progress condition implies at least $\lceil k/2 \rceil$ nonclean requests, so this proves the claim. $\square$

Lemma 6.4 (Universal menus). *Under the same slack condition, for each $1 \leq k \leq g$ there is a fixed menu of*

$$r_k = \left\lceil \frac{g(1 + 3 \log_2 N) + 1}{k} \right\rceil \quad (6)$$

direction k -tuples such that every set U described in Lemma 6.3 and every ordered list of k targets has a menu entry with at least $\lceil k/2 \rceil$ clean recovery sets.

Proof. A recovery set is specified by an anchor and a direction, giving at most ND_q descriptions. Allowing an unused marker in each of g slots, the number of occupied-set descriptions is at most $(1 + ND_q)^g$. The targets have at most N^k descriptions. Since $D_q \leq N$ and $k \leq g$, the total number of states is at most

$$(1 + ND_q)^g N^k \leq 2^{g(1+3 \log_2 N)}.$$

Counting line descriptions is essential: treating U as an arbitrary list of $g(q-1)$ points would introduce an extra factor of q in this exponent. Choose r_k candidate tuples independently. For a

fixed state, the probability that every tuple fails is at most 2^{-kr_k} by Lemma 6.3. A union bound over states is strictly below one by (6). Fix a menu working for all states as part of the circuit for parameters g, k, q, ℓ . $\square$

Theorem 6.5 (Nonuniform linear scheduling). *If $512gq \leq D_q$, a deterministic nonuniform Boolean circuit maps every ordered multiset of g targets in $\mathbb{F}_q^\ell$ to enumerated pairwise disjoint affine-line recovery sets using $\tilde{O}_\ell(gq)$ gates. More explicitly, its size is $O(gq\Lambda^5)$ for $\Lambda = \ell \log_2 q + \log_2(g + 2)$. The circuit's fixed menus have total description length $O(g(\log N)^2 \log(g + 2))$ bits.*

Proof. Maintain k active targets and the occupied set from the $g - k$ accepted requests. Initially $k = g$ and U is empty. Test the menu from Lemma 6.4, choose its first successful entry, and accept exactly the first $\lceil k/2 \rceil$ clean requests in that entry. These sets avoid U and each other. Compact the other $\lfloor k/2 \rfloor$ requests, retaining their original identifiers. This gives a fixed sequence of at most $1 + \lfloor \log_2 g \rfloor$ phase sizes, and never occupies more than g recovery sets. After the last phase, sort accepted records by original identifier to enumerate the answer sets in request order.

It remains to bound the cost of testing a menu. Generate records for all $r_k k(q - 1)$ candidate points, labelled by candidate, request, and line position, and append the $(g - k)(q - 1)$ occupied points *once*. Since $r_k k = O(g \log N)$, there are $O(gq \log N)$ records. Use these fixed passes:

- (i) Sort by point, with occupied records first at equal points. Along each equal-point run propagate an occupancy bit. In sorted position i , the update is $e_i = \text{occupied}_i \vee ((y_i = y_{i-1}) \wedge e_{i-1})$, with a dummy predecessor at the boundary. Thus every candidate point learns whether it lies in U .
- (ii) Compact candidate records and sort them by (candidate, y). Mark a record when its predecessor or successor has the same key. Points on one recovery line are distinct, so such a match is exactly an intersection between two requests in the same candidate.
- (iii) Sort by (candidate, request, line position) and OR the $q - 1$ bad-point flags for each request. Count the clean requests in each candidate and use priority selection to choose its first successful entry. Fixed multiplexers and compaction implement the acceptance step.

Every record has polynomial width in $\ell \log q$ and $\log(g + 2)$, and a constant number of sorting networks and linear scans suffices. Thus a phase has size $O(gq\Lambda^4)$: there are $O(gq\Lambda)$ records, with $O(\Lambda)$ bits per point record and $O(\Lambda^2)$ sorting-network comparators per record. Candidate directions and line scalars are hardwired, so generating a candidate point adds a constant field vector to the target using $O(\ell \log q)$ gates. Selecting and compacting whole line records costs at most $O(gq\Lambda^3)$ additional gates. There are $O(\Lambda)$ phases, proving the explicit bound. The menus contain $O(g \log N)$ directions per phase, each of $O(\log N)$ bits, giving the description bound. $\square$

6.3 Scatter and gather

The scheduler produces a sparse list of resource addresses, whereas the resource circuits occupy fixed positions in the circuit. *Scatter* sends each requested suffix to its resource position; *gather* returns the resulting value to the request that needs it. Sorting implements both maps with cost proportional, up to polynomial factors, to the lists being connected. With globally disjoint recovery sets, each resource position receives at most one suffix.

Every non-target point of a selected line produces an incidence record containing a request identifier, a line position, the point's code coordinate y , and the suffix to be evaluated. There are exactly

$t(q - 1) < tq$ such records, with distinct code coordinates.

The routing consists of four fixed sorting-and-local-update passes. Here “key” is the resource coordinate y ; any unlisted tie is broken by the record’s original position. The record-type order is pass-specific and is shown explicitly in the table.

Pass	Records	Sort order	Local result
Scatter match	incidence, slot	(key, type); incidence before slot	a matched slot takes its predecessor’s suffix; otherwise it takes a dummy
Scatter order	incidence, filled slot	(type, key); slots first	slots are in canonical key order
Gather match	value, saved incidence	(key, type); value before incidence	a matched incidence takes its predecessor’s value
Gather order	value, updated incidence	(type, request, line position); incidences first	decoder inputs are in canonical request order

Scatter records carry a d -bit suffix. After resource evaluation, value records carry the b -bit field value $G_y(z)$. Both payloads fit within the polynomial record-width bound.

Lemma 6.6 (Record routing). *Let a fixed table contain M distinct keys. Suppose every incidence key belongs to this table and occurs in at most one of the $t(q - 1)$ incidence records. With keys and payloads of width polynomial in n , $\ell \log q$, $\log(t + 2)$, and $\log(M + 2)$, suffixes can be scattered to their keyed slots and resource values gathered back to their request records using circuits of total size*

$$\tilde{O}_\ell(tq + M).$$

Implementation idea. The four passes in the table use a constant number of sorting networks on $O(tq + M)$ records of polynomial width. The exact fixed-wire treatment of unmatched records and array boundaries is given in [Appendix B.3](#).

For the resource bank above, the table consists of the $M = q^\ell$ field points.

7 The composition bound

Disjoint recovery lets each resource circuit serve at most one suffix. There are $q^\ell b$ Boolean resources, each on d variables, so ordinary one-copy synthesis bounds their total size by $q^\ell b L(d)$. The remaining cost comes from selecting recovery sets, routing suffixes and values, and XOR decoding.

Proposition 7.1 (Scheduler-sensitive composition). *Fix $\ell \geq 2$ and the local-recovery gadget for a split $n = p + d$. Suppose a Boolean circuit of size at most $S(t, q)$ maps every ordered list of t information points to enumerated pairwise disjoint affine-line recovery sets. Then every $f : \{0, 1\}^n \rightarrow \{0, 1\}$ satisfies*

$$C(f^{\times t}) \leq \underbrace{q^\ell b L(d)}_{\text{resource circuits}} + \underbrace{S(t, q)}_{\text{schedule selection}} + \underbrace{\tilde{O}_\ell(tq + q^\ell)}_{\text{routing and decoding}}. \quad (7)$$

Proof. Map each prefix to its information point and bit position, and apply the scheduler. This costs $\tilde{O}_\ell(t) + S(t, q)$ gates. By [Lemma 6.6](#), scatter the assigned suffixes to the q^ℓ resource positions,

filling unused positions with a fixed dummy suffix. Disjointness ensures that each position receives at most one requested suffix.

Evaluate the b Boolean functions $H_{y,r}$ at each position y , for a total of $q^\ell bL(d)$ gates, and gather the resulting field values. Routing costs $\tilde{O}_\ell(tq + q^\ell)$. For each request, XOR its $q - 1$ returned values and select the requested bit. This is correct by [Lemma 5.1](#) and costs $O(tqb)$, which the displayed overhead absorbs. $\square$

8 Direct proof of exponential-range mass production

The parameter choice balances storage against recovery capacity. Keeping $d \approx \delta n$ suffix bits leaves about $2^{(1-\delta)n}$ encoded prefix bits. Since a recovery set occupies about q points while a target has about $q^{\ell-1}$ directions, the scheduler needs t to be smaller than a constant multiple of $q^{\ell-2}$. The inequality below expresses exactly this requirement at the level of exponential rates.

Direct proof of [Theorem 1.1](#). Fix $\gamma < 1$, choose any fixed $0 < \delta < 1 - \gamma$, and put $d = \lceil \delta n \rceil$ and $p = n - d$. Choose a fixed $\ell \geq 2$ large enough that

$$\gamma < (1 - 2/\ell)(1 - \delta). \quad (8)$$

Use the product-code gadget with its minimal packing field. Its parameters satisfy

$$\log_2 q = (1 - \delta)n/\ell + o(n), \quad \log_2 D_q = (1 - 1/\ell)(1 - \delta)n + o(n).$$

Thus (8) implies $512tq \leq D_q$ for all sufficiently large n and every $t \leq 2^{\gamma n}$. Apply [Theorem 6.5](#) and [Proposition 7.1](#). This gives

$$C(f^{\times t}) \leq q^\ell bL(d) + \tilde{O}_\ell(tq + q^\ell).$$

The first term is $O_\ell(2^p 2^d/d) = O_\gamma(2^n/n)$. The two overhead exponents are at most $\gamma + (1 - \delta)/\ell < 1$ and $1 - \delta < 1$, respectively, with fixed positive margins. All suppressed polynomial factors are therefore negligible compared with $2^n/n$. Enlarge the constant to cover the finitely many initial lengths by ordinary circuit replication. $\square$

9 High-rate codes and the leading coefficient

The preceding proof has two constant losses. First, the product code's rate approaches $\ell^{-\ell}$: it stores substantially more symbols than the information it carries. Second, rounding up to the next allowed field size can leave a constant fraction of the information positions unused. We remove the first loss with a code of rate tending to one and the second with several smaller codewords, so that only the last one needs padding. The scheduler and decoder continue to use the same line-recovery interface.

Lifted Reed–Solomon codes already provide rate tending to one with line-parity recovery [[GKS13](#), [HPPV20](#)]. We give an elementary sufficient monomial construction below, followed by the packing and circuit-cost accounting.

Lemma 9.1 (High-rate line-parity code). *Fix integers $\ell \geq 2$ and $h \geq \lceil \log_2 \ell \rceil$. Put $q = 2^b$ and $N = q^\ell$. There is a linear code on $\mathbb{F}_q^\ell$ of dimension K such that*

$$\frac{K}{N} \geq 1 - (1 - 2^{-\ell h})^{\lfloor b/h \rfloor} = 1 - o_{\ell,h}(1) \quad (b \rightarrow \infty), \quad (9)$$

and each word P satisfies

$$P(u) = \sum_{\lambda \in \mathbb{F}_q^*} P(u + \lambda v)$$

for every target u and every nonzero direction v . The code can be made systematic on a fixed set of K evaluation points.

Proof. The low-degree condition in Lemma 5.1 was sufficient for line parity, but stronger than necessary. For the sum over $\mathbb{F}_q$ to vanish, it is enough that every positive degree appearing on a line avoid multiples of $q - 1$. We enforce this weaker condition using a common block of zero binary digits.

Consider reduced monomials $X_1^{e_1} \cdots X_\ell^{e_\ell}$ with $0 \leq e_i < 2^b$. Partition the first $h \lfloor b/h \rfloor$ binary digit positions into disjoint blocks of length h . Retain a monomial if some block is zero in *all* of its exponents. The fraction retained is exactly $1 - (1 - 2^{-\ell h})^{\lfloor b/h \rfloor}$. These monomials are linearly independent as evaluation functions, by repeated univariate interpolation.

To check line parity, expand a retained monomial at $u + \lambda v$. In characteristic two, every possibly nonzero term in this expansion has λ -degree $J = \sum_i j_i$, where the one-bits of j_i are a subset of those of e_i . This follows directly by writing $(u_i + \lambda v_i)^{e_i}$ as a product over the one-bits of e_i and using $(a + c)^{2^r} = a^{2^r} + c^{2^r}$.

Suppose the common zero block starts at position s . Every j_i has the same zero block, so its residue modulo 2^{s+h} is at most $2^s - 1$. Consequently,

$$J \bmod 2^{s+h} = \sum_i (j_i \bmod 2^{s+h}) \leq \ell(2^s - 1) \leq 2^{s+h} - \ell.$$

There is no wraparound in this equality. If J were a positive multiple $a(2^b - 1)$, we would have $1 \leq a \leq \ell - 1$: the zero block makes every $j_i < 2^b - 1$. Since $s + h \leq b$ and $a < 2^{s+h}$, its residue would instead be $2^{s+h} - a \geq 2^{s+h} - \ell + 1$, a contradiction. Thus every positive λ -degree is not divisible by $q - 1$. The monomial-sum calculation in Lemma 5.1 gives zero for the sum over $\mathbb{F}_q$, including the constant term. Linearity proves line parity for the span of the retained monomials.

Finally, choose K independent coordinate columns of a generator matrix. Evaluation on those coordinates is an isomorphism to $\mathbb{F}_q^K$, giving a systematic information set. The choice and any change of basis are offline parts of the nonuniform code specification. $\square$

To index the information set, store its ordered list of K points as a hardwired table, and sort its records together with all t live symbol indices, placing a table record before its queries at an equal index. Propagate the table value along each equal-index run, then sort queries back to request order. This costs $\tilde{O}_\ell(K + t)$ gates and handles repeated indices. It is the same batched lookup primitive used for occupancy broadcast, with a point as payload.

Proof of Theorem 1.2. Fix $0 < \delta < 1 - \gamma$, put $d = \lceil \delta n \rceil$, $p = n - d$, and choose a fixed ℓ satisfying (8). Fix $h = \lceil \log_2 \ell \rceil$ in Lemma 9.1. Instead of the minimal packing field, take

$$b = \left\lfloor \frac{p - 3 \log_2(n+2)}{\ell} \right\rfloor, \quad q = 2^b, \quad B = \left\lceil \frac{2^p}{Kb} \right\rceil,$$

for sufficiently large n , where K is the dimension in Lemma 9.1. Use B separate copies of this code. Each has Kb information bits, so they hold the 2^p prefix values with only the last code's information positions padded. Because $K/N \rightarrow 1$ and $Nb = o(2^p)$, their total number of Boolean resources is

$$2^p \leq BNb \leq \frac{N}{K} 2^p + Nb = (1 + o(1))2^p. \quad (10)$$

Here $B = \text{poly}_\ell(n)$ and $\log_2 q = (1 - \delta)n/\ell + O_\ell(\log n)$.

Fixed-divisor arithmetic maps each prefix to its code number, information index, and bit position. The batched table lookup above supplies all target points. Schedule these t targets together, even when they refer to different code copies. By Theorem 6.5, the resulting recovery sets are globally disjoint. Attaching the requested code number to each incidence therefore gives unique keys (code, y). Route the $t(q - 1)$ incidences against BN slots by the four-pass construction of Lemma 6.6.

For each of the BNb Boolean resource functions on d variables, use the sharp one-copy synthesis bound $L(d) \leq (1 + o(1))2^d/d$ in the stated basis [FM05]. The complete bound is

$$C(f^{\times t}) \leq BNbL(d) + \tilde{O}_\ell(tq + BN + K) \leq \left(\frac{1}{\delta} + o(1) \right) \frac{2^n}{n}.$$

Indeed, (10) controls the resource term; the other terms have exponents at most $\gamma + (1 - \delta)/\ell < 1$ and $1 - \delta < 1$. All estimates are uniform over f and the allowed t .

For every fixed $\delta < 1 - \gamma$ we have proved a limiting upper coefficient $1/\delta$. Taking the infimum of these bounds as δ increases to $1 - \gamma$ proves the theorem. This last step takes a limit of fixed-parameter bounds; it makes no assertion uniform in growing ℓ or in $\gamma \rightarrow 1$. $\square$

10 Optimal approximate quantum state mass production

We now prove Theorem 1.3. Its upper bound transfers Boolean mass production to quantum state preparation. Its lower bound counts how many states become distinguishable after taking tensor powers. Both depend on charging all elementary gates and measuring error on the entire batch.

10.1 Model and coherent Boolean evaluation

Fix the finite gate set $\mathcal{G} = \{H, T, T^\dagger, \text{CNOT}\}$, with arbitrary connectivity. Here H is the Hadamard gate, $T = \text{diag}(1, e^{i\pi/4})$, and CNOT is the controlled-NOT gate. Every gate has cost one. A synthesis circuit starts with all qubits in zero, has nt fixed designated output qubits, and returns every other qubit exactly to zero. The number of scratch qubits is unrestricted. Circuits have no measurements or postselection and may depend on the target state.

For normalized pure states, write

$$d(\psi, \phi) = \sqrt{1 - |\langle \psi | \phi \rangle|^2}$$

for their trace distance; this identifies states differing only by a global phase. Let $Q_\varepsilon(\psi, t)$ be the minimum gate count whose output is within trace distance ε of $|\psi\rangle^{\otimes t}$, and set

$$M(n, t) = \max_{\psi \in \mathbb{C}^{2^n}, \|\psi\|=1} Q_{1/10}(\psi, t).$$

The construction below bounds these integer-valued minima uniformly, so the maximum is well-defined. The state is specified when the circuit is designed; no copy of it is supplied as an input.

Lemma 10.1 (Coherent Boolean mass production). *If a Boolean circuit of size s computes $f^{\times t}$, there is a quantum circuit of size $O(s + t + 1)$ implementing*

$$|x_1, \dots, x_t\rangle |z_1, \dots, z_t\rangle |0^a\rangle \mapsto |x_1, \dots, x_t\rangle |z_1 \oplus f(x_1), \dots, z_t \oplus f(x_t)\rangle |0^a\rangle.$$

In particular, for fixed $0 \leq \eta < 1$, $k \geq 1$, and $t \leq 2^{\eta k}$, this map has size $O_\eta(2^k/k + t)$ for every k -bit Boolean function f .

Proof. Compute each Boolean gate into a fresh zero qubit, preserving its source qubits. NOT, AND, and OR each require a constant number of NOT, CNOT, and Toffoli gates; repeated sources and constants are handled by their one-input or constant simplifications. These gates have exact constant-size implementations over $\mathcal{G}$. Copy the t output bits to the z register with CNOTs, and reverse the computation. Prepared constant sources are also restored. Repeated use of a source as a control implements classical fan-out without an additional gate per use beyond those already charged for computing its successors.

The displayed equality holds on every computational basis state, hence by linearity on every superposition and on inputs entangled with other registers. Apply [Theorem 1.1](#) for the final assertion. $\square$

10.2 Conditional rotations with clean scratch space

For $a \in \{y, z\}$, let $R_a(\theta) = \exp(-i\theta\sigma_a/2)$, where σ_a is the corresponding Pauli matrix. A *multiplexed rotation* on k address qubits and one target qubit is

$$U_\theta = \sum_{x \in \{0,1\}^k} |x\rangle\langle x| \otimes R_a(\theta(x)).$$

Thus an address selects a rotation angle while remaining unchanged. We use $\|\cdot\|$ for the operator norm on operators.

Lemma 10.2 (Batched conditional rotations). *Fix $0 \leq \eta < 1$. Suppose $k \geq 1$, $t \leq 2^{\eta k}$, and $0 < \delta \leq 1/4$. Put $b = \lceil \log_2(8\pi/\delta) \rceil$. For every multiplexed rotation U_θ , a circuit over $\mathcal{G}$ with clean scratch space approximates $U_\theta^{\otimes t}$ to operator-norm error at most $t\delta$, using*

$$O_\eta\left(b \frac{2^k}{k} + t \text{ poly}(b)\right)$$

gates. The polynomial is fixed, independent of k, t, θ . The circuit preserves every address qubit exactly.

Proof. Represent each angle modulo 4π , and round it down to $4\pi \sum_{j=1}^b 2^{-j} f_j(x)$, where each f_j is a Boolean function of the address. Since $\|R_a(u) - R_a(v)\| \leq |u - v|/2$, this changes each selected rotation by at most $2\pi 2^{-b} \leq \delta/4$.

For each j , use [Lemma 10.1](#) to compute $f_j(x_1), \dots, f_j(x_t)$ into zero qubits. On block i , rotate the target by $4\pi 2^{-j}$ controlled by the computed bit, and then uncompute the bits. This charges $O_\eta(b2^k/k + bt)$ gates for Boolean evaluation and its reversal. The address registers and all computed control bits are unchanged by the controlled rotations, so reversing the lookup erases its scratch space exactly.

It remains to synthesize a controlled rotation in the finite gate set without losing this property. Write X for the NOT matrix. Given a one-qubit word A approximating $R_a(\theta/2)$, use the two-CNOT circuit with block form

$$|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes AXA^\dagger X.$$

The identity $XR_a(u)X = R_a(-u)$ shows that the ideal choice of A implements the desired controlled rotation. For every implemented word A , the control-zero block is exactly I and the control is preserved exactly. A global phase in A cancels against $A^\dagger$. The Solovay–Kitaev theorem [\[DN06\]](#) supplies A of length polynomial in $\log(b/\delta)$ so that this controlled gate has error at most $\delta/(4b)$.

After lookup and uncomputation the resulting operation on the data is the tensor product of the corresponding approximate one-block multiplexors: each computed bit depends only on its own block’s address. The error in one block is at most $\delta/4 + b\delta/(4b) \leq \delta$. The tensor-product telescoping inequality gives error at most $t\delta$. There are bt synthesized controlled gates. Since $\log(b/\delta) = O(b)$, their total cost is $t \text{poly}(b)$, as claimed. $\square$

Lemma 10.3 (State preparation by conditional rotations). *Up to global phase, every n -qubit pure state can be prepared using two multiplexed rotations, one about each of the y and z axes, at each address length $k = 0, \dots, n-1$.*

Proof. This is the standard recursive state-synthesis decomposition [\[SBM06, Kre23\]](#). To see it directly, write the target as

$$|\psi\rangle = \sum_{x \in \{0,1\}^{n-1}} |x\rangle (a_x|0\rangle + b_x|1\rangle).$$

For each x , choose angles θ_x, ϕ_x and a complex scalar c_x such that

$$a_x|0\rangle + b_x|1\rangle = c_x R_z(\phi_x) R_y(\theta_x) |0\rangle, \quad |c_x|^2 = |a_x|^2 + |b_x|^2.$$

Two rotations parametrize any normalized one-qubit state up to phase; when both amplitudes vanish take $c_x = 0$ and arbitrary angles. The state $\sum_x c_x |x\rangle$ is normalized. Prepare it recursively, adjoin a zero target, and apply the two rotations controlled by x . The induction ends with a scalar of modulus one, which contributes only an irrelevant global phase. $\square$

10.3 Upper bound and precision allocation

Proposition 10.4 (Quantum upper bound). *For every fixed $0 \leq \gamma < 1$, every n -qubit pure state ψ , and every $1 \leq t \leq 2^{\gamma n}$,*

$$Q_{1/10}(\psi, t) = O_\gamma \left(\frac{2^n}{n} \log_2(t+1) \right).$$

Proof. Fix constants $\gamma < \alpha < 1$ and $\gamma/\alpha < \eta < 1$, and put $m = \lceil \alpha n \rceil$ and $L = 1 + \log_2 t$. For sufficiently large n we have $1 \leq m < n$. Use [Lemma 10.3](#) on each of the t blocks. At address length k , approximate each of the two rotations with the one-block error allowance

$$\delta_k = \frac{1}{40t} 2^{k-n}.$$

The number of angle bits in [Lemma 10.2](#) is then

$$b_k = O(n - k + L) = O(n).$$

Since $t\delta_k$ bounds the error of each batched rotation, telescoping over the stages gives total operator-norm error at most

$$2t \sum_{k=0}^{n-1} \delta_k = \frac{1}{20} \sum_{k=0}^{n-1} 2^{k-n} < \frac{1}{10}.$$

Acting on the all-zero state gives the same bound on vector distance after aligning global phases, and hence on trace distance.

To apply the Boolean theorem at short address lengths, replace k by $k' = \max\{k, m\}$. When $k < m$, add zero address qubits and extend each angle function to ignore them. They are preserved and can be reused. The fixed rate η works at every stage, since

$$\log_2 t \leq \gamma n < \eta \alpha n \leq \eta k'.$$

Thus no stage invokes a mass-production bound outside its allowed range.

For $k \geq m$, [Lemma 10.2](#) bounds the Boolean lookup cost, summed over the two rotations per stage, by

$$\begin{aligned} O_\gamma \left(\sum_{k=m}^{n-1} (n - k + L) \frac{2^k}{k} \right) &\leq O_\gamma \left(\frac{2^n}{n} \sum_{j \geq 1} (j + L) 2^{-j} \right) \\ &= O_\gamma \left(\frac{2^n}{n} L \right). \end{aligned}$$

Here $k \geq \alpha n$ bounds the denominators, and the two geometric sums are finite. The $O(n)$ short stages each have $b_k = O(n)$ and address length m , for a total lookup cost

$$O_\gamma(n^2 2^m / m) = O_\gamma(n 2^{\alpha n}) = o(2^n / n).$$

Finally, all rotation-synthesis and output-copying costs sum to $t \text{poly}(n) = o(2^n / n)$ uniformly for $t \leq 2^{\gamma n}$. The strict fixed margins $\alpha < 1$ and $\gamma < 1$ justify both absorptions. The scratch space is restored exactly at each stage. Enlarging the constant covers the finitely many remaining input lengths. $\square$

The geometric precision allocation is useful even for one copy: the largest address tables need only $O(L)$ precision bits, while the extra precision at short tables has a geometrically smaller synthesis cost. Related allocations appear in [\[GKW26, Appendix B\]](#).

10.4 Packing tensor powers and counting circuits

Lemma 10.5 (Packing tensor-power states). *For $D \geq 2$ and $t \geq 1$, there is a set $\mathcal{P}$ of pure states in $\mathbb{C}^D$ with $|\mathcal{P}| \geq (4t)^{D-1}$ such that distinct $\psi, \phi \in \mathcal{P}$ satisfy*

$$d(\psi^{\otimes t}, \phi^{\otimes t}) \geq \sqrt{1 - e^{-1/4}} > 2/10.$$

Proof. For a fixed pure state ψ and a Haar-random pure state ϕ , a trace-distance ball of radius $r \in [0, 1]$ has probability

$$\Pr[d(\psi, \phi) \leq r] = r^{2(D-1)}.$$

Indeed, unitary invariance lets us take ψ to be the first basis vector. Normalize D independent standard complex Gaussian entries to generate ϕ . Their squared moduli are independent exponentials, and the first divided by their sum has the Beta(1, $D - 1$) distribution. Its upper-tail probability at $1 - r^2$ is $(r^2)^{D-1}$.

Choose a maximal set of states separated by at least $r = 1/(2\sqrt{t})$. Compactness guarantees such a finite set, and maximality makes its radius- r balls cover the pure-state space. Their volumes give $|\mathcal{P}|r^{2(D-1)} \geq 1$. For distinct members,

$$\begin{aligned} d(\psi^{\otimes t}, \phi^{\otimes t})^2 &= 1 - |\langle \psi | \phi \rangle|^{2t} \\ &\geq 1 - \left(1 - \frac{1}{4t}\right)^t \geq 1 - e^{-1/4}. \end{aligned}$$

This proves both assertions. □

Proposition 10.6 (Quantum lower bound). *Uniformly over integers $1 \leq t \leq 2^n$ as $n \rightarrow \infty$,*

$$M(n, t) = \Omega\left(nt + \frac{2^n}{n} \log_2(t + 1)\right).$$

Proof. Put $D = 2^n$ and use [Lemma 10.5](#). The triangle inequality implies that one circuit output can approximate at most one of the $|\mathcal{P}|$ tensor-power targets to error $1/10$.

Let $g \geq 1$ bound the number of gates. A circuit uses at most $2g$ scratch qubits that participate in any gate; unused zero qubits can be removed. Canonically relabel the scratch qubits by their first use, keeping the nt designated output wires fixed. A gate is specified by its type in the fixed finite set $\mathcal{G}$ and at most two wire indices in a set of size $nt + 2g$. Therefore the number of circuits of size at most g is at most

$$(g + 1)(|\mathcal{G}|(nt + 2g + 2)^2)^g.$$

This upper bound includes circuits that fail to restore their scratch space, so imposing the clean-scratch condition cannot invalidate it.

If every target had a circuit of size at most g , taking logarithms and comparing with $|\mathcal{P}|$ would give

$$(D - 1) \log_2(4t) \leq O(g \log_2(nt + g + 2) + \log_2(g + 1)).$$

For $g \leq 2^n$ and $t \leq 2^n$, the logarithm on the right is $O(n)$, which yields the logarithmic term. If $g > 2^n$, that term already follows from $\log_2(t + 1) \leq n + 1$.

For the remaining nt term, take $|\psi\rangle = |1^n\rangle$. Any untouched output wire stays in zero, making the output orthogonal to $|1^{nt}\rangle$. Error below one therefore requires every output wire to participate in a

gate, and a gate touches at most two wires. Thus $M(n, t) \geq nt/2$. Combining the two lower bounds proves the claim. $\square$

[Propositions 10.4](#) and [10.6](#) prove [Theorem 1.3](#). The lower bound is a worst-case existence statement; it does not identify an explicit family of hard states. Together the bounds determine the order of growth, not its leading constant, and do not assert an efficient uniform procedure for finding the circuits.

11 Implications and open problems

The Boolean results leave questions about efficient construction of the menus, the optimal leading coefficient, and copy rates approaching one. The quantum application determines the fixed-rate order of growth, but also leaves a transition near the endpoint.

Can the menus be constructed efficiently? [Lemma 6.4](#) proves that short menus exist simultaneously for all inputs. Exhaustive search is a finite construction, but no efficient uniform procedure for finding such menus is supplied.

What is the optimal leading coefficient? The normalized worst-case size has lower limiting coefficient at least 1 by one-copy counting, and [Theorem 1.2](#) gives an upper limiting coefficient $1/(1 - \gamma)$. In particular, preserving coefficient 1 at a positive fixed exponential copy rate remains unresolved here.

Where does the plateau end? The theorem reaches $2^{(1-\varepsilon)n}$ copies for every fixed $\varepsilon > 0$, but does not treat a rate that approaches one with n . Some degradation in that regime is forced. Suppose $n \geq 2$ and f depends on all n of its variables. Restrict a circuit for $f^{\times t}$ to the union of its output cones, and write s for the remaining gate count. Every one of the tn inputs remains. Each connected component of the underlying undirected graph contains an output, so there are at most t components. The graph therefore has at least $tn + s - t$ edges, while fan-in at most two permits at most $2s$ edges. Hence

$$C(f^{\times t}) \geq t(n - 1).$$

A bound of the form $A 2^n/n$ therefore cannot hold once $t > A 2^n/(n(n - 1))$; in particular, a uniform $O(2^n/n)$ plateau must fail for $t = \omega(2^n/n^2)$. The theorem still leaves the detailed mass-production curve throughout the near-endpoint regime $t = 2^{n-o(n)}$ unresolved. Analyzing this regime requires uniform control of the code rate and circuit overhead as the parameters vary with n .

What happens near the quantum endpoint? For $t \leq 2^n$, [Proposition 10.6](#) gives $M(n, t) = \Omega(nt + (2^n/n) \log_2(t + 1))$. [Theorem 1.3](#) matches this order for each fixed rate below one, where nt is negligible, but does not settle $t = 2^{n-o(n)}$. In particular, the $\Theta(2^n)$ cost at fixed positive rates cannot continue to $t = 2^n$, where touching all output wires already costs $\Omega(n2^n)$. Matching the two-term lower bound near this endpoint, or optimizing scratch space and T count simultaneously with total gate count, remains open here. The state theorem also does not settle the corresponding optimal tradeoff for arbitrary unitary operations.

A An explicit recursive construction

The explicitly constructible greedy scheduler gives a recursive proof of [Theorem 1.1](#). The leading-coefficient bound in [Theorem 1.2](#) uses the menu scheduler from the main text. Apply the composition bound recursively. At level k , split the input into $k + 1$ approximately equal blocks, encode one block, and invoke the preceding level on the remaining k .

For $r \geq 1$, define

$$L_r(n) = \max_{f: \{0,1\}^n \rightarrow \{0,1\}} C(f^{\times r}).$$

For each f , choose a circuit for $f^{\times r}$ of size at most $L_r(n)$. If a construction supplies fewer than r live inputs, the unused inputs of this circuit may be fixed arbitrarily and their outputs ignored. Thus a bound for $L_r(n)$ also applies to every smaller number of copies.

A.1 Grouping and composition

Partition the t requests into C fixed groups, each of size at most

$$g = \left\lceil \frac{t}{C} \right\rceil.$$

Apply [Lemma 6.2](#) independently inside each group. A code coordinate is used at most once per group and therefore at most C times overall. The inequalities $\lceil t/C \rceil \leq t/C + 1$ and $2t \leq t^2/C + C$, the latter being the arithmetic–geometric mean inequality, show that the total scheduling cost is

$$C \cdot \tilde{O}_\ell(g^2 q) = \tilde{O}_\ell\left(\frac{t^2 q}{C} + Cq\right). \quad (11)$$

The construction is valid whenever (5) holds for $g = \lceil t/C \rceil$, that is, whenever

$$\left\lceil \frac{t}{C} \right\rceil q < Dq. \quad (12)$$

Increasing C reduces scheduling cost and increases the number of suffixes each resource must handle. The recursion chooses C to balance these costs.

Bounded-demand invariant. Fix f and the local-recovery gadget above. Each $H_{y,r}$ is one fixed Boolean function on d variables. Inside each request group, disjoint recovery ensures that coordinate y occurs at most once; hence the pair (y, r) receives at most one suffix from each of the C groups. Place those suffixes in the canonical group slots and pad unused slots with a fixed dummy suffix. One circuit for $H_{y,r}^{\times C}$, of size at most $L_C(d)$, serves exactly this bounded demand. There are $q^\ell b$ such pairs, and no sharing between different pairs is assumed.

Proposition A.1 (Grouped composition). *Fix $\ell \geq 2$ and the local-recovery gadget for a split $n = p + d$. Let $C \geq 1$, set $g = \lceil t/C \rceil$, and suppose that, for every group size $1 \leq h \leq g$, a Boolean circuit of size at most $S(g, q)$ maps every multiset of h information points to enumerated pairwise disjoint affine-line recovery sets. Then every $f : \{0, 1\}^n \rightarrow \{0, 1\}$ satisfies*

$$C(f^{\times t}) \leq \underbrace{q^\ell b L_C(d)}_{\text{resource circuits}} + \underbrace{CS(g, q)}_{\text{schedule selection}} + \underbrace{\tilde{O}_\ell(tq + Cq^\ell)}_{\text{routing and decoding}}. \quad (13)$$

Proof. Apply the assembly from [Proposition 7.1](#), with one scheduler per group at total cost $CS(g, q)$. Use [Lemma 6.6](#) with $M = Cq^\ell$ keys (group, y). After scatter, fixed wiring transposes the canonical key order to (y, group) , placing all suffixes for one resource together.

The bounded-demand invariant permits evaluating $H_{y,r}^{\times C}$ in each resource position, at total cost $q^\ell b L_C(d)$. Gather and decode as in [Proposition 7.1](#). Prefix mapping, routing, and decoding together cost $\tilde{O}_\ell(tq + Cq^\ell)$. $\square$

Proposition A.2 (Composition bound). *Fix $\ell \geq 2$. Let $n = p + d$ with p sufficiently large depending on ℓ , choose $q = 2^b$ minimally so that $s_0^\ell b \geq 2^p$, and suppose [Equation \(12\)](#) holds. Then every $f : \{0, 1\}^n \rightarrow \{0, 1\}$ satisfies*

$$C(f^{\times t}) \leq \underbrace{q^\ell b L_C(d)}_{\text{recursive resource evaluation}} + \tilde{O}_\ell \left(\underbrace{\frac{t^2 q}{C}}_{\text{line scheduling}} + \underbrace{tq}_{\text{incidences and decoding}} + \underbrace{Cq^\ell}_{\text{resource slots}} \right). \quad (14)$$

The induction keeps the resource term at scale $2^n/n$ while making all three overhead terms exponentially smaller.

Proof. Put $g = \lceil t/C \rceil$. The scheduling condition lets us apply [Lemma 6.2](#) with $S(g, q) = \tilde{O}_\ell(g^2 q)$. Moreover,

$$Cg^2 q \leq C \left(\frac{t}{C} + 1 \right)^2 q = O \left(\frac{t^2 q}{C} + tq + Cq \right).$$

Apply [Proposition A.1](#); the tq term is already present, and Cq is absorbed by Cq^ℓ because $q \geq 2$ and $\ell \geq 2$. $\square$

A.2 The induction invariant

For an integer $k \geq 1$, let P_k denote the assertion that, for every fixed $0 \leq \gamma < k/(k+1)$, there is a constant $A_{k,\gamma}$ such that, for all sufficiently large n , every $f : \{0, 1\}^n \rightarrow \{0, 1\}$, and every integer $1 \leq t \leq 2^{\gamma n}$,

$$C(f^{\times t}) \leq A_{k,\gamma} \frac{2^n}{n}.$$

Every comparison below has a fixed positive linear margin in its base-two logarithm. We use without further comment that such a margin absorbs the $o(n)$ terms, every fixed polynomial factor hidden by $\tilde{O}$, and the $O(1)$ changes caused by floors and ceilings.

A.3 Two equal blocks: the base case

Lemma A.3 (Two-block base). P_1 holds.

Here $C = 1$, so no mass-production hypothesis is used inside a resource. The two equal blocks make the resource term proportional to $2^{n/2} L(n/2)$; the one-half threshold comes from the quadratic scheduler term $t^2 q$.

Proof. Fix $0 \leq \gamma < 1/2$, let $\eta = 1/2 - \gamma > 0$, and suppose $t \leq 2^{\gamma n}$. Put

$$m = \left\lfloor \frac{n}{2} \right\rfloor, \quad p = m, \quad d = n - m, \quad C = 1.$$

Thus $n = 2m + O(1)$ and all recovery sets will be globally disjoint. In block units, the request and code parameters are

$$\log_2 t \leq m - \eta n + O(1), \quad \log_2 q = \frac{m}{\ell} + o(n), \quad \log_2 D_q = \left(1 - \frac{1}{\ell}\right) m + o(n).$$

Choose a sufficiently large fixed ℓ so that

$$\frac{1}{\ell} < \eta. \tag{15}$$

Since $2m/\ell \leq n/\ell < \eta n$, this gives

$$m - \eta n + \frac{m}{\ell} < m - \frac{m}{\ell}.$$

The strict linear margin now implies $tq < D_q$ for all sufficiently large n .

The resource term in Equation (14) is, by Shannon–Lupanov synthesis,

$$q^\ell bL(d) = O_\ell \left(2^p \frac{2^d}{d} \right) = O_\gamma(2^n/n).$$

The scheduler exponent is

$$\log_2(t^2 q) \leq 2m - 2\eta n + \frac{m}{\ell} + o(n) < n,$$

where the strict inequality follows from (15). Likewise, $\log_2(tq) \leq m - \eta n + m/\ell + o(n) < n$ and $\log_2 q^\ell = m + o(n) < n$. After the suppressed polynomial factors are restored, every overhead term is $2^{(1-\Omega_\gamma(1))n}$ and is negligible compared with $2^n/n$. The same strict margins absorb all rounding. $\square$

A.4 Adding one equal block

Lemma A.4 (Equal-block step). *For every integer $k \geq 2$, P_{k-1} implies P_k .*

Proof. Fix $0 \leq \gamma < k/(k+1)$. If $\gamma < (k-1)/k$, the assertion is already P_{k-1} . We may therefore assume

$$\frac{k-1}{k} \leq \gamma < \frac{k}{k+1}.$$

Let

$$\eta = \frac{k}{k+1} - \gamma > 0, \quad m = \left\lfloor \frac{n}{k+1} \right\rfloor, \quad p = m, \quad d = n - m.$$

Write $n = (k+1)m + r$ with $0 \leq r \leq k$, and choose the number of request groups to be

$$C = \left\lceil 2^{(k-1)m - \eta n/2} \right\rceil. \tag{16}$$

This choice splits the available slack in half. Dividing the requests among C groups puts each group $\eta n/2$ below the one-block exponent m ; measured on the remaining d bits, the exponent of C is also

a fixed amount below the threshold supplied by P_{k-1} . The assumption $\gamma \geq (k-1)/k$ ensures that the exponent in (16) is positive for all sufficiently large n .

Let $\beta = (k-1)/k$. Since $d = km + r$,

$$\beta d - \log_2 C = \frac{\eta n}{2} + \frac{k-1}{k}r + O(1).$$

In particular, $C \leq 2^{(\beta-\eta/4)d}$ for all sufficiently large n . The exponent $\beta - \eta/4$ is fixed and strictly below the threshold for P_{k-1} , so the inductive hypothesis gives

$$L_C(d) = O_{k,\gamma}(2^d/d).$$

By Equation (1), the resource term is therefore

$$q^\ell b L_C(d) = O_{k,\gamma,\ell}\left(2^m \frac{2^d}{d}\right) = O_{k,\gamma,\ell}(2^n/n).$$

The remaining estimates are easiest to read in block units. Since $\log_2 t \leq \gamma n = km - \eta n + O_k(1)$, the size $g = \lceil t/C \rceil$ of a request group satisfies

$$\log_2 g \leq m - \frac{\eta n}{2} + O_k(1).$$

The code parameters satisfy

$$\log_2 q = \frac{m}{\ell} + o(n), \quad \log_2 D_q = \left(1 - \frac{1}{\ell}\right)m + o(n).$$

Choose a sufficiently large fixed ℓ so that

$$\frac{2}{(k+1)\ell} < \frac{\eta}{2}. \tag{17}$$

Then $2m/\ell < \eta n/2$, and hence

$$m - \frac{\eta n}{2} + \frac{m}{\ell} < m - \frac{m}{\ell}.$$

Together with (3), this is exactly the strict exponent margin required for Equation (12).

Finally, the complete overhead ledger in Equation (14) is

$$\begin{aligned} \log_2 \left(\frac{t^2 q}{C} \right) &\leq (k+1)m - \frac{3\eta n}{2} + \frac{m}{\ell} + o(n) < n, \\ \log_2(tq) &\leq n - m - \eta n + \frac{m}{\ell} + o(n) < n, \\ \log_2(Cq^\ell) &\leq n - m - \frac{\eta n}{2} + o(n) < n. \end{aligned}$$

The first strict inequality uses $(k+1)m \leq n$ and (17); the other two use $m = \Theta_k(n)$ and $\ell > 1$. Thus every overhead term is exponentially smaller than $2^n/n$. This proves P_k . $\square$

Alternative proof of Theorem 1.1. By Lemmas A.3 and A.4, P_k holds for every $k \geq 1$. Fix $0 \leq \gamma < 1$ and choose k so that $\gamma < k/(k+1)$. Then P_k gives the asserted bound beyond a threshold N_γ . For the finitely many integers $1 \leq n < N_\gamma$, use $C(f^{\times t}) \leq tC(f)$ and enlarge A_γ to dominate $ntC(f)/2^n$ over all relevant n , f , and t . This proves the statement for every $n \geq 1$. $\square$

B Circuit implementation details

This appendix records the gate-level details behind the scheduler and routing lemmas. The main proof uses only their statements and costs.

B.1 Projective-direction indexing

Proof of Lemma 6.1. Normalize a nonzero vector so that its first nonzero coordinate is one. The normalized directions whose first nonzero coordinate is in position $h \in \{1, \dots, \ell\}$ have the form

$$(0, \dots, 0, 1, w_{h+1}, \dots, w_\ell)$$

and form a block of size $q^{\ell-h}$. Let

$$B_h = \sum_{r=1}^{h-1} q^{\ell-r}, \quad \text{val}_q(w_{h+1}, \dots, w_\ell) = \sum_{r=h+1}^{\ell} \text{int}(w_r) q^{\ell-r},$$

where $\text{int}(w_r) \in [q]_0$ is the canonical binary representation. The rank is

$$B_h + \text{val}_q(w_{h+1}, \dots, w_\ell).$$

The blocks are consecutive and their total size is D_q .

To rank, find the first nonzero coordinate, invert it, normalize the vector, and concatenate the b -bit tail coordinates. To unrank, compare the rank with the ℓ hardwired block intervals, subtract the appropriate B_h , and split the remainder into base- q digits. Since $q = 2^b$, the last step is bit slicing. The field operations and comparisons from the circuit conventions give size $\text{poly}_\ell(b)$ in both directions. $\square$

B.2 The least-missing-direction circuit

Proof of Lemma 6.2. At step j , fewer than D_q directions are forbidden, so a valid direction exists. This proves the combinatorial statement.

For the circuit bound, represent a projective direction by normalizing its first nonzero coordinate to one and use the ranking from Lemma 6.1. At stage j , for each $y \in U$, output the sentinel rank D_q when $y = u_j$ and otherwise output the projective rank of $y - u_j$. At the gate level, first multiplex a fixed nonzero vector in place of $y - u_j$ when $y = u_j$, rank that always-nonzero vector, and then overwrite the result by D_q on the equality branch. Use $\lceil \log_2(D_q + 1) \rceil$ bits per rank. There are $(j-1)(q-1) = O(jq)$ such entries.

Sort these ranks as $r_1 \leq \dots \leq r_s$. The least missing rank can be found without a dense D_q -bit table. Rank zero is a candidate if $s = 0$ or $r_1 > 0$; after position i , rank $r_i + 1$ is a candidate if

$$r_i < D_q - 1 \quad \text{and} \quad (i = s \text{ or } r_{i+1} > r_i + 1).$$

These tests are applied to the full sorted list, including duplicates and sentinels. Only the last occurrence of a non-sentinel rank can generate its successor, and a sentinel cannot generate a candidate.

A left-to-right prefix-OR circuit identifies the first candidate in this order. Multiplexers select its rank using $O(s \log D_q)$ gates.

Since $|U \setminus \{u_j\}| < D_q$, fewer than D_q distinct non-sentinel ranks are forbidden, so a candidate exists. Sorting, comparisons, and priority selection therefore cost $\tilde{O}_\ell(jq)$ gates. Unrank the selected direction using [Lemma 6.1](#) and enumerate its $q-1$ non-target line points, at cost $q \text{poly}_\ell(b)$. Unrolling the stages and summing over $j \leq g$ gives $\tilde{O}_\ell(g^2q)$. $\square$

B.3 Four-pass fixed-wire routing

Proof of [Lemma 6.6](#). For scatter, create one slot record for each of the M table keys and combine them with the fewer than tq incidence records. Preserve a copy of every incidence record by free fan-out. The uniqueness hypothesis puts at most one incidence at each key. After the first pass in the table, a slot therefore copies the preceding suffix exactly when a type-and-key equality test succeeds; otherwise it receives a fixed dummy suffix. The predecessor of the first position is defined to be a dummy record, so the boundary case uses the same guard. The second pass places the filled slots in canonical key order, ready for the resource circuits.

For gather, combine the preserved incidences with one value record per key. The third pass places that value immediately before the unique matching incidence; a guarded local multiplexer copies it into the incidence. The fourth pass puts the first $t(q-1)$ positions in the fixed request-and-line order required by the decoders. Thus absent incidences, the first array position, and all type boundaries are handled by fixed dummy data and the same guarded update, not by data-dependent wiring.

This is a constant number of sorting networks on $O(tq + M)$ records. Each record has width $\text{poly}_\ell(n, \log q, \log(t+2), \log(M+2))$, so the standard primitives from the circuit conventions give the stated bound. $\square$

Relation to multiselection. The gather half is also an instance of multiselection. Equivalently, it is a partial-function inner join in the sense of Holmgren and Rothblum [[HR24](#), Proposition 25 and Sections 6.1–6.2]. Write M for the number of value slots and $R = t(q-1)$ for the number of incidences. Uniqueness of incidence keys gives $R \leq M$. For the resource tables used here, a key’s field coordinates and any group or code number determine its slot index by fixed arithmetic. Each incidence in canonical request–line–position order supplies this index; applying their binary main theorem independently to the b bit positions therefore gives size $O(bM + bR \log^3(M+2))$ [[HR24](#), Main theorem], within the bound of [Lemma 6.6](#). The four-pass construction handles scatter and gather within a single routing bound.

C Formalization and exact finite accounting

This appendix connects the asymptotic proof in [Section 9](#) with the Lean theorem. It records how the formal construction tracks requests, counts resource circuits, and handles arbitrary input lengths using exact integer parameters.

Companion and reproducibility. At revision [8dd82c9](#), the Lean 4.33.1/mathlib 4.33.1 module `Algebraic.MassProduction` exposes `BlockInduction.exponentialMassProduction` for the ex-

PLICIT proof and `Nonuniform.realSharpMassProduction` for the nonuniform proof. Both endpoints use only Lean’s standard axioms, with no proof placeholders. A small executable Python model of the menu verifier and the high-rate subcode, with exhaustive checks on tiny parameters, accompanies the manuscript source at revision [1c880d9](#).

Request identity and inactive slots. Each request carries a distinct fixed identifier, even when its target and suffix coincide with another request’s. The scheduler maintains a permutation of accepted and pending requests, together with disjoint recovery lines for the accepted prefix. Each phase preserves all original payloads. For regular array sizes, a line uses q scalar slots: the zero scalar is marked inactive, and its recovered Boolean contribution is fixed to zero. After XOR recovery, only the identifier and the one-bit answer need to be sorted back into request order. Thus restoring output order does not require another permutation of the full point lists.

Charge exactly the resource bank. A single hardwired table indexed by the source prefix can return the code number, information point, and basis-bit selector together. Batched lookup handles repeated prefixes and removes runtime division from this version of the construction. The resource key is (code number, point, basis bit); valid scheduled incidences have distinct keys. Evaluate precisely the $R = BNb$ Boolean resource circuits, once each. Padding the sorting networks adds inactive routing records, not resource evaluations. This distinction avoids a constant-factor loss in the main term. If $\widehat{L}(d)$ bounds the chosen one-copy synthesis circuits, the full cost is bounded by

$$R\widehat{L}(d) + H, \quad H \leq c_\ell 2^p n^7$$

in the parameter regime below, with c_ℓ independent of n, f, t . For each fixed positive integer Q , the actual synthesis circuits eventually admit $\widehat{L}(d) = \lfloor (Q+1)2^d/(Qd) \rfloor$. This checked polynomial envelope is coarser than the optimized pass count in [Theorem 6.5](#); both give the required negligible overhead.

Whole field blocks, arbitrary input lengths. One exact parameter choice fixes positive integers h, ℓ, G, S , with $\ell \leq 2^h$, and sets

$$P = \ell h + 1 < S, \quad m = \lfloor n/S \rfloor, \quad b = hm, \quad p = Pm, \quad d = n - p.$$

Choose $G + h + 1 \leq h(\ell - 1)$ and $\gamma < G/S$. For sufficiently large m , the scheduler batch 2^{Gm} covers the requested copies, including rounding, and has $512 2^{Gm} q \leq D_q$. Also $2^{Gm} q \leq 2^p$. Put the incomplete input block into the suffix; then

$$p + d = n, \quad \frac{d}{n} \geq \frac{S - P}{S} > 0.$$

Integer slopes can make P/S arbitrarily close to γ from above while keeping all these inequalities strict. All dimensions and slopes are fixed before n varies.

For $A = 2^{\ell h}$ the retained dimension on this subsequence is exactly $K = A^m - (A - 1)^m$. At a fixed positive integer precision Q , the elementary rate estimate is

$$m \geq Q(A - 1) \implies QA^m \leq (Q + 1)K.$$

The formal packing uses $B = \lfloor 2^p/(Kb) \rfloor + 1$, charging at most one extra codeword. Since $Nb = 2^{\ell h m} hm$ and $p = (\ell h + 1)m$, eventually $QNb \leq 2^p$. Consequently $QR \leq (Q + 2)2^p$. The number B

need not be polynomial in this alternative parameter choice; the bounds on the total resource bank and the key widths are what matter. Finally, $m \leq d$ and every fixed polynomial in n is eventually smaller than 2^m . These facts absorb H after normalization by $n/2^n$.

Uniform quantifiers. For a rational rate $u/v < 1$ and every positive integer precision J , the checked integer endpoint supplies one cutoff, uniform over f and all positive $t \leq 2^{\lfloor un/v \rfloor}$, with

$$J(v - u)n C(f^{\times t}) \leq (J + 1)v2^n.$$

The final real-rate theorem chooses a slightly larger rational rate and a large enough precision inside the requested additive error. Upward exponent rounding is covered by the scheduler’s strict capacity margin. The result has exactly the real-rate and ε quantifiers of [Theorem 1.2](#), without substituting growing dimensions into a fixed-dimension estimate.

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AI Disclosure

This project began in an interactive conversation on ChatGPT.com that was exported to the author’s computer as a L^AT_EX document. Later interactive sessions used Claude Fable 5 and 5.1 through Claude Code and ChatGPT 5.6 Sol through Codex. These systems materially influenced the mathematical experiments, broad literature searches, proof strategy, and internal constructions, including the scheduling and recursive-composition approach; their role went beyond prose editing. The author selected the directions to pursue, checked and revised the proofs and literature claims, and takes full responsibility for the correctness, originality, and integrity of the paper. Codex also assisted with development and testing of the nonuniform menu scheduler and the high-rate-code refinement, and with their Lean formalization through the sharp real-rate theorem. Codex also assisted with the quantum state preparation application and its packing lower bound.

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